

CHAPTER VII

SURVEY RESEARCH

Shortly after the terrorist attacks in New York City and Washington, DC, in September of 2001, researcher Jennifer Lerner and her colleagues conducted an Internet-based survey of nearly 2,000 American teens and adults ranging in age from 13 to 88 (Lerner, Gonzalez, Small, & Fischhoff, 2003)¹. They asked participants about their reactions to the attacks and for their judgments of various terrorism-related and other risks. Among the results were that the participants tended to overestimate most risks, that females did so more than males, and that there were no differences between teens and adults. The most interesting result, however, had to do with the fact that some participants were “primed” to feel anger by asking them what made them angry about the attacks and by presenting them with a photograph and audio clip intended to evoke anger. Others were primed to feel fear by asking them what made them fearful about the attacks and by presenting them with a photograph and audio clip intended to evoke fear. As the researchers hypothesized, the participants who were primed to feel anger perceived less risk than the participants who had been primed to feel fear—showing how risk perceptions are strongly tied to specific emotions.

The study by Lerner and her colleagues is an example of survey research in psychology—the topic of this chapter. We begin with an overview of survey research, including its definition, some history, and a bit about who conducts it and why. We then look at survey responding as a psychological process and the implications of this for constructing good survey questionnaires. Finally, we consider some issues related to actually conducting survey research, including sampling the participants and collecting the data.

Notes

1. Lerner, J. S., Gonzalez, R. M., Small, D. A., & Fischhoff, B. (2003). Effects of fear and anger on perceived risks of terrorism: A national field experiment. *Psychological Science*, 14, 144–150.

34. Overview of Survey Research

Learning Objectives

1. Define what survey research is, including its two important characteristics.
2. Describe several different ways that survey research can be used and give some examples.

What Is Survey Research?

Survey research is a quantitative and qualitative method with two important characteristics. First, the variables of interest are measured using self-reports (using questionnaires or interviews). In essence, survey researchers ask their participants (who are often called **respondents** in survey research) to report directly on their own thoughts, feelings, and behaviors. Second, considerable attention is paid to the issue of sampling. In particular, survey researchers have a strong preference for large random samples because they provide the most accurate estimates of what is true in the population. In fact, survey research may be the only approach in psychology in which random sampling is routinely used. Beyond these two characteristics, almost anything goes in survey research. Surveys can be long or short. They can be conducted in person, by telephone, through the mail, or over the Internet. They can be about voting intentions, consumer preferences, social attitudes, health, or anything else that it is possible to ask people about and receive meaningful answers. Although survey data are often analyzed using statistics, there are many questions that lend themselves to more qualitative analysis.

Most survey research is non-experimental. It is used to describe single variables (e.g., the percentage of voters who prefer one presidential candidate or another, the prevalence of schizophrenia in the general population, etc.) and also to assess statistical relationships between variables (e.g., the relationship between income and health). But surveys can also be used within experimental research. The study by Lerner and her colleagues is a good example. Their use of self-report measures and a large national sample identifies their work as survey research. But their manipulation of an independent variable (anger vs. fear) to assess its effect on a dependent variable (risk judgments) also identifies their work as experimental.

History and Uses of Survey Research

Survey research may have its roots in English and American “social surveys” conducted around the turn of the 20th century by researchers and reformers who wanted to document the extent of social problems such as poverty (Converse, 1987)¹. By the 1930s, the US government was conducting surveys to document

economic and social conditions in the country. The need to draw conclusions about the entire population helped spur advances in sampling procedures. At about the same time, several researchers who had already made a name for themselves in market research, studying consumer preferences for American businesses, turned their attention to election polling. A watershed event was the presidential election of 1936 between Alf Landon and Franklin Roosevelt. A magazine called *Literary Digest* conducted a survey by sending ballots (which were also subscription requests) to millions of Americans. Based on this “straw poll,” the editors predicted that Landon would win in a landslide. At the same time, the new pollsters were using scientific methods with much smaller samples to predict just the opposite—that Roosevelt would win in a landslide. In fact, one of them, George Gallup, publicly criticized the methods of *Literary Digest* before the election and all but guaranteed that his prediction would be correct. And of course, it was, demonstrating the effectiveness of careful survey methodology (We will consider the reasons that Gallup was right later in this chapter). Gallup’s demonstration of the power of careful survey methods led later researchers to local, and in 1948, the first national election survey by the Survey Research Center at the University of Michigan. This work eventually became the American National Election Studies (<https://electionstudies.org/>) as a collaboration of Stanford University and the University of Michigan, and these studies continue today.

From market research and election polling, survey research made its way into several academic fields, including political science, sociology, and public health—where it continues to be one of the primary approaches to collecting new data. Beginning in the 1930s, psychologists made important advances in questionnaire design, including techniques that are still used today, such as the Likert scale. (See “What Is a Likert Scale?” in Section 7.2 “Constructing Survey Questionnaires”.) Survey research has a strong historical association with the social psychological study of attitudes, stereotypes, and prejudice. Early attitude researchers were also among the first psychologists to seek larger and more diverse samples than the convenience samples of university students that were routinely used in psychology (and still are).

Survey research continues to be important in psychology today. For example, survey data have been instrumental in estimating the prevalence of various mental disorders and identifying statistical relationships among those disorders and with various other factors. The National Comorbidity Survey is a large-scale mental health survey conducted in the United States (see <http://www.hcp.med.harvard.edu/ncs>). In just one part of this survey, nearly 10,000 adults were given a structured mental health interview in their homes in 2002 and 2003. Table 7.1 presents results on the lifetime prevalence of some anxiety, mood, and substance use disorders. (Lifetime prevalence is the percentage of the population that develops the problem sometime in their lifetime.) Obviously, this kind of information can be of great use both to basic researchers seeking to understand the causes and correlates of mental disorders as well as to clinicians and policymakers who need to understand exactly how common these disorders are.

Table 7.1 Some Lifetime Prevalence Results From the National Comorbidity Survey

Lifetime prevalence*			
Disorder	Total	Female	Male
Generalized anxiety disorder	5.7	7.1	4.2
Obsessive-compulsive disorder	2.3	3.1	1.6
Major depressive disorder	16.9	20.2	13.2
Bipolar disorder	4.4	4.5	4.3
Alcohol abuse	13.2	7.5	19.6
Drug abuse	8.0	4.8	11.6
*The lifetime prevalence of a disorder is the percentage of people in the population that develop that disorder at any time in their lives.			

And as the opening example makes clear, survey research can even be used as a data collection method within experimental research to test specific hypotheses about causal relationships between variables. Such studies, when conducted on large and diverse samples, can be a useful supplement to laboratory studies conducted on university students. Survey research is thus a flexible approach that can be used to study a variety of basic and applied research questions.

Notes

1. Converse, J. M. (1987). *Survey research in the United States: Roots and emergence, 1890–1960*. Berkeley, CA: University of California Press.

35. Constructing Surveys

Learning Objectives

1. Describe the cognitive processes involved in responding to a survey item.
2. Explain what a context effect is and give some examples.
3. Create a simple survey questionnaire based on principles of effective item writing and organization.

The heart of any survey research project is the survey itself. Although it is easy to think of interesting questions to ask people, constructing a good survey is not easy at all. The problem is that the answers people give can be influenced in unintended ways by the wording of the items, the order of the items, the response options provided, and many other factors. At best, these influences add noise to the data. At worst, they result in systematic biases and misleading results. In this section, therefore, we consider some principles for constructing surveys to minimize these unintended effects and thereby maximize the reliability and validity of respondents' answers.

Survey Responding as a Psychological Process

Before looking at specific principles of survey construction, it will help to consider survey responding as a psychological process.

A Cognitive Model

Figure 7.1 presents a model of the cognitive processes that people engage in when responding to a survey item (Sudman, Bradburn, & Schwarz, 1996)¹. Respondents must interpret the question, retrieve relevant information from memory, form a tentative judgment, convert the tentative judgment into one of the response options provided (e.g., a rating on a 1-to-7 scale), and finally edit their response as necessary.



Figure 7.1 Model of the Cognitive Processes Involved in Responding to a Survey Item. [\[Image description\]](#)

Consider, for example, the following questionnaire item:

How many alcoholic drinks do you consume in a typical day?

- _____ a lot more than average
- _____ somewhat more than average
- _____ average
- _____ somewhat fewer than average
- _____ a lot fewer than average

Although this item at first seems straightforward, it poses several difficulties for respondents. First, they must interpret the question. For example, they must decide whether “alcoholic drinks” include beer and wine (as opposed to just hard liquor) and whether a “typical day” is a typical weekday, typical weekend day, or both. Even though Chang and Krosnick (2003)² found that asking about “typical” behavior has been shown to be more valid than asking about “past” behavior, their study compared “typical week” to “past week” and may be different when considering typical weekdays or weekend days). Once respondents have interpreted the question, they must retrieve relevant information from memory to answer it. But what information should they retrieve, and how should they go about retrieving it? They might think vaguely about some recent occasions on which they drank alcohol, they might carefully try to recall and count the number of alcoholic drinks they consumed last week, or they might retrieve some existing beliefs that they have about themselves (e.g., “I am not much of a drinker”). Then they must use this information to arrive at a tentative judgment about how many alcoholic drinks they consume in a typical day. For example, this mental calculation might mean dividing the number of alcoholic drinks they consumed last week by seven to come up with an average number per day. Then they must format this tentative answer in terms of the response options actually provided. In this case, the options pose additional problems of interpretation. For example, what does “average” mean, and what would count as “somewhat more” than average? Finally, they must decide whether they want to report the response they have come up with or whether they want to edit it in some way. For example, if they believe that they drink a lot more than average, they might not want to report that for fear of looking bad in the eyes of the researcher, so instead, they may opt to select the “somewhat more than average” response option.

From this perspective, what at first appears to be a simple matter of asking people how much they drink (and receiving a straightforward answer from them) turns out to be much more complex.

Context Effects on Survey Responses

Again, this complexity can lead to unintended influences on respondents' answers. These are often referred to as **context effects** because they are not related to the content of the item but to the context in which the item appears (Schwarz & Strack, 1990)³. For example, there is an **item-order effect** when the order in which the items are presented affects people's responses. One item can change how participants interpret a later item or change the information that they retrieve to respond to later items. For example, researcher Fritz Strack and his colleagues asked college students about both their general life satisfaction and their dating frequency (Strack, Martin, & Schwarz, 1988)⁴. When the life satisfaction item came first, the correlation between the two was only $-.12$, suggesting that the two variables are only weakly related. But when the dating frequency item came first, the correlation between the two was $+.66$, suggesting that those who date more have a strong tendency to be more satisfied with their lives. Reporting the dating frequency first made that information more accessible in memory so that they were more likely to base their life satisfaction rating on it.

The response options provided can also have unintended effects on people's responses (Schwarz, 1999)⁵. For example, when people are asked how often they are "really irritated" and given response options ranging from "less than once a year" to "more than once a month," they tend to think of major irritations and report being irritated infrequently. But when they are given response options ranging from "less than once a day" to "several times a month," they tend to think of minor irritations and report being irritated frequently. People also tend to assume that middle response options represent what is normal or typical. So if they think of themselves as normal or typical, they tend to choose middle response options. For example, people are likely to report watching more television when the response options are centered on a middle option of 4 hours than when centered on a middle option of 2 hours. To mitigate against order effects, rotate questions and response items when there is no natural order. Counterbalancing or randomizing the order of presentation of the questions in online surveys are good practices for survey questions and can reduce response order effects that show that among undecided voters, the first candidate listed in a ballot receives a 2.5% boost simply by virtue of being listed first⁶!

Writing Survey Items

Types of Items

Questionnaire items can be either open-ended or closed-ended. **Open-ended items** simply ask a question

and allow participants to answer in whatever way they choose. The following are examples of open-ended questionnaire items.

- “What is the most important thing to teach children to prepare them for life?”
- “Please describe a time when you were discriminated against because of your age.”
- “Is there anything else you would like to tell us about?”

Open-ended items are useful when researchers do not know how participants might respond or when they want to avoid influencing their responses. Open-ended items are more qualitative in nature, so they tend to be used when researchers have more vaguely defined research questions—often in the early stages of a research project. Open-ended items are relatively easy to write because there are no response options to worry about. However, they take more time and effort on the part of participants, and they are more difficult for the researcher to analyze because the answers must be transcribed, coded, and submitted to some form of qualitative analysis, such as content analysis. Another disadvantage is that respondents are more likely to skip open-ended items because they take longer to answer. It is best to use open-ended questions when the answer is unsure or for quantities which can easily be converted to categories later in the analysis.

Closed-ended items ask a question and provide a set of response options for participants to choose from. The alcohol item just mentioned is an example, as are the following:

How old are you?

- _____ Under 18
- _____ 18 to 34
- _____ 35 to 49
- _____ 50 to 70
- _____ Over 70

On a scale of 0 (no pain at all) to 10 (worst pain ever experienced), how much pain are you in right now?

Have you ever in your adult life been depressed for a period of 2 weeks or more? Yes No

Closed-ended items are used when researchers have a good idea of the different responses that participants might make. They are more quantitative in nature, so they are also used when researchers are interested in a well-defined variable or construct such as participants' level of agreement with some statement, perceptions of risk, or frequency of a particular behavior. Closed-ended items are more difficult to write because they must include an appropriate set of response options. However, they are relatively quick and easy for participants to complete. They are also much easier for researchers to analyze because the responses can be easily converted to numbers and entered into a spreadsheet. For these reasons, closed-ended items are much more common.

All closed-ended items include a set of response options from which a participant must choose. For

categorical variables like sex, race, or political party preference, the categories are usually listed and participants choose the one (or ones) to which they belong. For quantitative variables, a rating scale is typically provided. A **rating scale** is an ordered set of responses that participants must choose from. Figure 7.2 shows several examples. The number of response options on a typical rating scale ranges from three to 11—although five and seven are probably most common. Five-point scales are best for unipolar scales where only one construct is tested, such as frequency (Never, Rarely, Sometimes, Often, Always). Seven-point scales are best for bipolar scales where there is a dichotomous spectrum, such as liking (Like very much, Like somewhat, Like slightly, Neither like nor dislike, Dislike slightly, Dislike somewhat, Dislike very much). For bipolar questions, it is useful to offer an earlier question that branches them into an area of the scale; if asking about liking ice cream, first ask “Do you generally like or dislike ice cream?” Once the respondent chooses like or dislike, refine it by offering them relevant choices from the seven-point scale. Branching improves both reliability and validity (Krosnick & Berent, 1993)⁷. Although you often see scales with numerical labels, it is best to only present verbal labels to the respondents but convert them to numerical values in the analyses. Avoid partial labels or length or overly specific labels. In some cases, the verbal labels can be supplemented with (or even replaced by) meaningful graphics. The last rating scale shown in Figure 7.2 is a visual-analog scale, on which participants make a mark somewhere along the horizontal line to indicate the magnitude of their response.

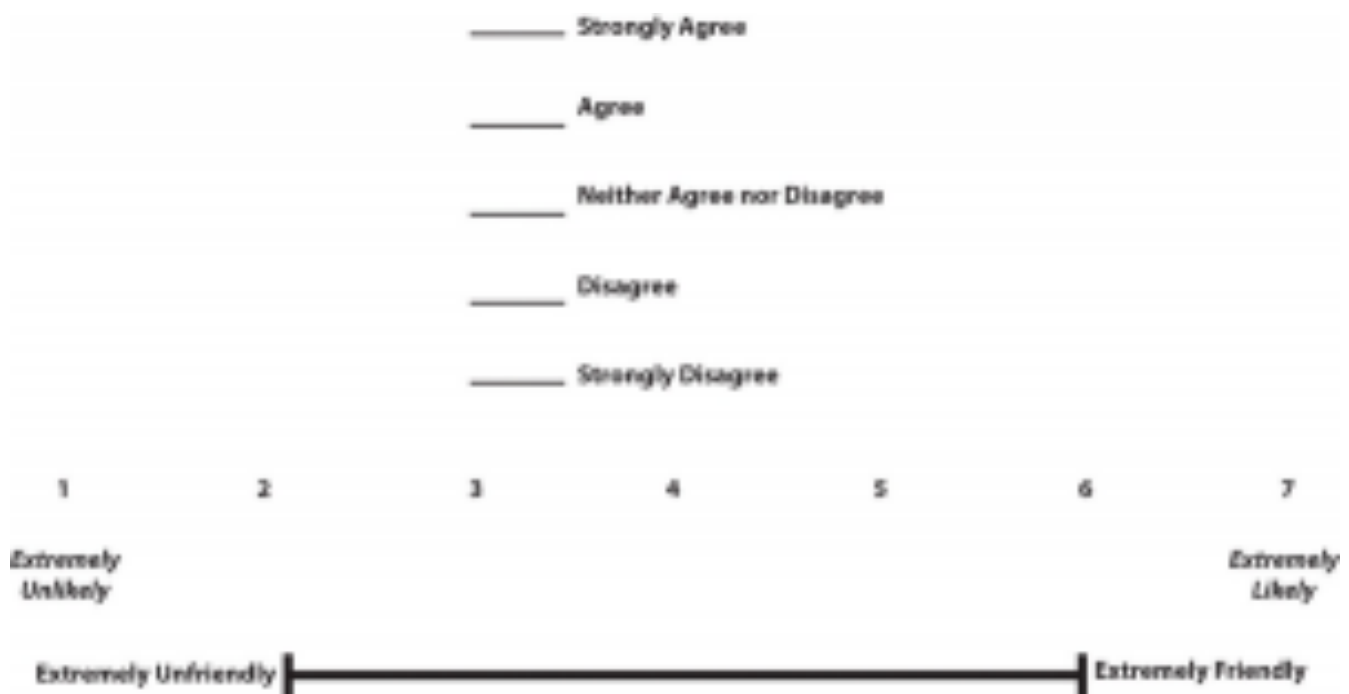


Figure 7.2 Example Rating Scales for Closed-Ended Questionnaire Items. [\[Image description\]](#)

What Is a Likert Scale?

In reading about psychological research, you are likely to encounter the term *Likert scale*. Although this term is sometimes used to refer to almost any rating scale (e.g., a 0-to-10 life satisfaction scale), it has a much more precise meaning.

In the 1930s, researcher Rensis Likert (pronounced LICK-ert) created a new approach for measuring people's attitudes (Likert, 1932)⁸. It involves presenting people with several statements—including both favorable and unfavorable statements—about some person, group, or idea. Respondents then express their agreement or disagreement with each statement on a 5-point scale: *Strongly Agree*, *Agree*, *Neither Agree nor Disagree*, *Disagree*, *Strongly Disagree*. Numbers are assigned to each response and then summed across all items to produce a score representing the attitude toward the person, group, or idea. For items that are phrased in an opposite direction (e.g., negatively worded statements instead of positively worded statements), reverse coding is used so that the numerical scoring of statements also runs in the opposite direction. The entire set of items came to be called a Likert scale.

Thus unless you are measuring people's attitude toward something by assessing their level of agreement with several statements about it, it is best to avoid calling it a Likert scale. You are probably just using a "rating scale."

Writing Effective Items

We can now consider some principles of writing questionnaire items that minimize unintended context effects and maximize the reliability and validity of participants' responses. A rough guideline for writing questionnaire items is provided by the BRUSO model (Peterson, 2000)⁹. An acronym, **BRUSO** stands for "brief," "relevant," "unambiguous," "specific," and "objective." Effective questionnaire items are *brief* and to the point. They avoid long, overly technical, or unnecessary words. This brevity makes them easier for respondents to understand and faster for them to complete. Effective questionnaire items are also *relevant* to the research question. If a respondent's sexual orientation, marital status, or income is not relevant, then items on them should probably not be included. Again, this makes the questionnaire faster to complete, but it also avoids annoying respondents with what they will rightly perceive as irrelevant or even "nosy" questions. Effective questionnaire items are also *unambiguous*; they can be interpreted in only one way. Part of the problem with the alcohol item presented earlier in this section is that different respondents might have different ideas about what constitutes "an alcoholic drink" or "a typical day." Effective questionnaire items are also *specific* so that it is clear to respondents what their response *should* be about and clear to researchers what it is about. A common problem here is closed-ended items that are "double barreled." They ask about two conceptually separate issues but allow only one response. For example, "Please rate the extent to which you have been feeling anxious and depressed." This item should probably be split into two separate items—one about anxiety and one about depression. Finally, effective questionnaire items are *objective* in the sense that they do not reveal the researcher's own opinions

or lead participants to answer in a particular way. Table 7.2 shows some examples of poor and effective questionnaire items based on the BRUSO criteria. The best way to know how people interpret the wording of the question is to conduct a pilot test and ask a few people to explain how they interpreted the question.

Table 7.2 BRUSO Model of Writing Effective Questionnaire Items, Plus Examples

Criterion	Poor	Effective
B—Brief	“Are you now or have you ever been the possessor of a firearm?”	“Have you ever owned a gun?”
R—Relevant	“What is your sexual orientation?”	Do not include this item unless it is clearly relevant to the research.
U—Unambiguous	“Are you a gun person?”	“Do you currently own a gun?”
S—Specific	“How much have you read about the new gun control measure and sales tax?”	“How much have you read about the new sales tax?”
O—Objective	“How much do you support the new gun control measure?”	“What is your view of the new gun control measure?”

For closed-ended items, it is also important to create an appropriate response scale. For categorical variables, the categories presented should generally be mutually exclusive and exhaustive. Mutually exclusive categories do not overlap. For a religion item, for example, the categories of *Christian* and *Catholic* are not mutually exclusive but *Protestant* and *Catholic* are mutually exclusive. Exhaustive categories cover all possible responses. Although *Protestant* and *Catholic* are mutually exclusive, they are not exhaustive because there are many other religious categories that a respondent might select: *Jewish*, *Hindu*, *Buddhist*, and so on. In many cases, it is not feasible to include every possible category, in which case an *Other* category, with a space for the respondent to fill in a more specific response, is a good solution. If respondents could belong to more than one category (e.g., race), they should be instructed to choose all categories that apply.

For rating scales, five or seven response options generally allow about as much precision as respondents are capable of. However, numerical scales with more options can sometimes be appropriate. For dimensions such as attractiveness, pain, and likelihood, a 0-to-10 scale will be familiar to many respondents and easy for them to use. Regardless of the number of response options, the most extreme ones should generally be “balanced” around a neutral or modal midpoint. An example of an unbalanced rating scale measuring perceived likelihood might look like this:

Unlikely | Somewhat Likely | Likely | Very Likely | Extremely Likely

A balanced version might look like this:

Extremely Unlikely | Somewhat Unlikely | As Likely as Not | Somewhat Likely |Extremely Likely

Note, however, that a middle or neutral response option does not have to be included. Researchers sometimes choose to leave it out because they want to encourage respondents to think more deeply about

their response and not simply choose the middle option by default. However, including middle alternatives on bipolar dimensions can be used to allow people to choose an option that is neither.



Figure 7.3 “Question” retrieved from <http://imgs.xkcd.com/comics/question.png> (CC-BY-NC 2.5). [\[Image description\]](#)

Formatting the Survey

Writing effective items is only one part of constructing a survey. For one thing, every survey should have a written or spoken introduction that serves two basic functions (Peterson, 2000)¹⁰. One is to encourage respondents to participate in the survey. In many types of research, such encouragement is not necessary either because participants do not know they are in a study (as in naturalistic observation) or because they are part of a subject pool and have already shown their willingness to participate by signing up and showing up for the study. Survey research usually catches respondents by surprise when they answer their phone, go to their mailbox, or check their e-mail—and the researcher must make a good case for why they should agree to participate. Thus the introduction should briefly explain the purpose of the survey and its importance, provide information about the sponsor of the survey (university-based surveys tend to generate higher response rates), acknowledge the importance of the respondent’s participation, and describe any incentives for participating.

The second function of the introduction is to establish informed consent. Remember that this involves describing to respondents everything that might affect their decision to participate. This includes the topics covered by the survey, the amount of time it is likely to take, the respondent’s option to withdraw at any time, confidentiality issues, and so on. Written consent forms are not always used in survey research (when the research is of minimal risk and completion of the survey instrument is often accepted by the IRB as evidence of consent to participate), so it is important that this part of the introduction be well documented and presented clearly and in its entirety to every respondent.

The introduction should be followed by the substantive questionnaire items. But first, it is important to

present clear instructions for completing the questionnaire, including examples of how to use any unusual response scales. Remember that the introduction is the point at which respondents are usually most interested and least fatigued, so it is good practice to start with the most important items for purposes of the research and proceed to less important items. Items should also be grouped by topic or by type. For example, items using the same rating scale (e.g., a 5-point agreement scale) should be grouped together if possible to make things faster and easier for respondents. Demographic items are often presented last because they are least interesting to participants but also easy to answer in the event respondents have become tired or bored. Of course, any survey should end with an expression of appreciation to the respondent.

Image Descriptions

Figure 7.1 image description: Flowchart modelling the cognitive processes involved in responding to a survey item. In order, these processes are:

- Question Interpretation
- Information Retrieval
- Judgment Formation
- Response Formatting
- Response Editing

[\[Return to Figure 7.1\]](#)

Figure 7.2 image description: Three different rating scales for survey questions. The first scale provides a choice between “strongly agree,” “agree,” “neither agree nor disagree,” “disagree,” and “strongly disagree.” The second is a scale from 1 to 7, with 1 being “extremely unlikely” and 7 being “extremely likely.” The third is a sliding scale, with one end marked “extremely unfriendly” and the other “extremely friendly.” [\[Return to Figure 7.2\]](#)

Figure 7.3 image description: A note reads, “Dear Isaac. Do you like me?” with two check boxes reading “yes” or “no.” Someone has added a third check box, which they’ve checked, that reads, “There is as yet insufficient data for a meaningful answer.” [\[Return to Figure 7.3\]](#)

Media Attributions

- [Question](#) by XKCD license under [CC BY-NC 2.5](#)

Notes

1. Sudman, S., Bradburn, N. M., & Schwarz, N. (1996). *Thinking about answers: The application of cognitive processes to survey methodology*. San Francisco, CA: Jossey-Bass.
2. Chang, L., & Krosnick, J.A. (2003). Measuring the frequency of regular behaviors: Comparing the 'typical week' to the 'past week'. *Sociological Methodology*, 33, 55-80.
3. Schwarz, N., & Strack, F. (1990). Context effects in attitude surveys: Applying cognitive theory to social research. In W. Stroebe & M. Hewstone (Eds.), *European review of social psychology* (Vol. 2, pp. 31-50). Chichester, UK: Wiley.
4. Strack, F., Martin, L. L., & Schwarz, N. (1988). Priming and communication: The social determinants of information use in judgments of life satisfaction. *European Journal of Social Psychology*, 18, 429-442.
5. Schwarz, N. (1999). Self-reports: How the questions shape the answers. *American Psychologist*, 54, 93-105.
6. Miller, J.M. & Krosnick, J.A. (1998). The impact of candidate name order on election outcomes. *Public Opinion Quarterly*, 62(3), 291-330.
7. Krosnick, J.A. & Berent, M.K. (1993). Comparisons of party identification and policy preferences: The impact of survey question format. *American Journal of Political Science*, 27(3), 941-964.
8. Likert, R. (1932). A technique for the measurement of attitudes. *Archives of Psychology*, 140, 1-55.
9. Peterson, R. A. (2000). *Constructing effective questionnaires*. Thousand Oaks, CA: Sage.
10. Peterson, R. A. (2000). *Constructing effective questionnaires*. Thousand Oaks, CA: Sage.

36. Conducting Surveys

Learning Objectives

1. Explain the difference between probability and non-probability sampling, and describe the major types of probability sampling.
2. Define sampling bias in general and non-response bias in particular. List some techniques that can be used to increase the response rate and reduce non-response bias.
3. List the four major ways to conduct a survey along with some pros and cons of each.

In this section, we consider how to go about conducting a survey. We first consider the issue of sampling, followed by some different methods of actually collecting survey data.

Sampling

Essentially all psychological research involves sampling—selecting a sample to study from the population of interest. Sampling falls into two broad categories. The first category, **Probability sampling**, occurs when the researcher can specify the probability that each member of the population will be selected for the sample. The second is **Non-probability sampling**, which occurs when the researcher cannot specify these probabilities. Most psychological research involves non-probability sampling. For example, **Convenience sampling**—studying individuals who happen to be nearby and willing to participate—is a very common form of non-probability sampling used in psychological research. Other forms of non-probability sampling include **snowball sampling** (in which existing research participants help recruit additional participants for the study), **quota sampling** (in which subgroups in the sample are recruited to be proportional to those subgroups in the population), and **self-selection sampling** (in which individuals choose to take part in the research on their own accord, without being approached by the researcher directly).

Survey researchers, however, are much more likely to use some form of probability sampling. This tendency is because the goal of most survey research is to make accurate estimates about what is true in a particular population, and these estimates are most accurate when based on a probability sample. For example, it is important for survey researchers to base their estimates of election outcomes—which are often decided by only a few percentage points—on probability samples of likely registered voters.

Compared with non-probability sampling, probability sampling requires a very clear specification of the population, which of course depends on the research questions to be answered. The population might be all registered voters in Washington State, all American consumers who have purchased a car in the

past year, women in the Seattle over 40 years old who have received a mammogram in the past decade, or all the alumni of a particular university. Once the population has been specified, probability sampling requires a **sampling frame**. This sampling frame is essentially a list of all the members of the population from which to select the respondents. Sampling frames can come from a variety of sources, including telephone directories, lists of registered voters, and hospital or insurance records. In some cases, a map can serve as a sampling frame, allowing for the selection of cities, streets, or households.

There are a variety of different probability sampling methods. **Simple random sampling** is done in such a way that each individual in the population has an equal probability of being selected for the sample. This type of sampling could involve putting the names of all individuals in the sampling frame into a hat, mixing them up, and then drawing out the number needed for the sample. Given that most sampling frames take the form of computer files, random sampling is more likely to involve computerized sorting or selection of respondents. A common approach in telephone surveys is random-digit dialing, in which a computer randomly generates phone numbers from among the possible phone numbers within a given geographic area.

A common alternative to simple random sampling is **stratified random sampling**, in which the population is divided into different subgroups or “strata” (usually based on demographic characteristics) and then a random sample is taken from each “stratum.” **Proportionate stratified random sampling** can be used to select a sample in which the proportion of respondents in each of various subgroups matches the proportion in the population. For example, because about 12.6% of the American population is African American, stratified random sampling can be used to ensure that a survey of 1,000 American adults includes about 126 African-American respondents. **Disproportionate stratified random sampling** can also be used to sample extra respondents from particularly small subgroups—allowing valid conclusions to be drawn about those subgroups. For example, because Asian Americans make up a relatively small percentage of the American population (about 5.6%), a simple random sample of 1,000 American adults might include too few Asian Americans to draw any conclusions about them as distinct from any other subgroup. If representation is important to the research question, however, then disproportionate stratified random sampling could be used to ensure that enough Asian-American respondents are included in the sample to draw valid conclusions about Asian Americans as a whole.

Yet another type of probability sampling is **cluster sampling**, in which larger clusters of individuals are randomly sampled and then individuals within each cluster are randomly sampled. This is the only probability sampling method that does not require a sampling frame. For example, to select a sample of small-town residents in Washington, a researcher might randomly select several small towns and then randomly select several individuals within each town. Cluster sampling is especially useful for surveys that involve face-to-face interviewing because it minimizes the amount of traveling that the interviewers must do. For example, instead of traveling to 200 small towns to interview 200 residents, a research team could travel to 10 small towns and interview 20 residents of each. The National Comorbidity Survey was done using a form of cluster sampling.

How large does a survey sample need to be? In general, this estimate depends on two factors. One is the level of confidence in the result that the researcher wants. The larger the sample, the closer any statistic based on that sample will tend to be to the corresponding value in the population. The other factor is a practical constraint in the form of the budget of the study. Larger samples provide greater confidence,

but they take more time, effort, and money to obtain. Taking these two factors into account, most survey research uses sample sizes that range from about 100 to about 1,000. Conducting a power analysis prior to launching the survey helps to guide the researcher in making this trade-off.

Sample Size and Population Size

Why is a sample of about 1,000 considered to be adequate for most survey research—even when the population is much larger than that? Consider, for example, that a sample of only 1,000 American adults is generally considered a good sample of the roughly 252 million adults in the American population—even though it includes only about 0.000004% of the population! The answer is a bit surprising.

One part of the answer is that a statistic based on a larger sample will tend to be closer to the population value and that this can be characterized mathematically. Imagine, for example, that in a sample of registered voters, exactly 50% say they intend to vote for the incumbent. If there are 100 voters in this sample, then there is a 95% chance that the true percentage in the population is between 40 and 60. But if there are 1,000 voters in the sample, then there is a 95% chance that the true percentage in the population is between 47 and 53. Although this “95% confidence interval” continues to shrink as the sample size increases, it does so at a slower rate. For example, if there are 2,000 voters in the sample, then this reduction only reduces the 95% confidence interval to 48 to 52. In many situations, the small increase in confidence beyond a sample size of 1,000 is not considered to be worth the additional time, effort, and money.

Another part of the answer—and perhaps the more surprising part—is that confidence intervals depend only on the size of the sample and not on the size of the population. So a sample of 1,000 would produce a 95% confidence interval of 47 to 53 regardless of whether the population size was a hundred thousand, a million, or a hundred million.

Sampling Bias

Probability sampling was developed in large part to address the issue of sampling bias. **Sampling bias** occurs when a sample is selected in such a way that it is not representative of the entire population and therefore produces inaccurate results. This bias was the reason that the *Literary Digest* straw poll was so far off in its prediction of the 1936 presidential election. The mailing lists used came largely from telephone directories and lists of registered automobile owners, which over-represented wealthier people, who were more likely to vote for Landon. Gallup was successful because he knew about this bias and found ways to sample less wealthy people as well.

There is one form of sampling bias that even careful random sampling is subject to. It is almost never the case that everyone selected for the sample actually responds to the survey. Some may have died or moved away, and others may decline to participate because they are too busy, are not interested in the

survey topic, or do not participate in surveys on principle. If these survey non-responders differ from survey responders in systematic ways, then this difference can produce **non-response bias**. For example, in a mail survey on alcohol consumption, researcher Vivienne Lahaut and colleagues found that only about half the sample responded after the initial contact and two follow-up reminders (Lahaut, Jansen, van de Mheen, & Garretsen, 2002)¹. The danger here is that the half who responded might have different patterns of alcohol consumption than the half who did not, which could lead to inaccurate conclusions on the part of the researchers. So to test for non-response bias, the researchers later made unannounced visits to the homes of a subset of the non-responders—coming back up to five times if they did not find them at home. They found that the original non-responders included an especially high proportion of abstainers (nondrinkers), which meant that their estimates of alcohol consumption based only on the original responders were too high.

Although there are methods for statistically correcting for non-response bias, they are based on assumptions about the non-responders—for example, that they are more similar to late responders than to early responders—which may not be correct. For this reason, the best approach to minimizing non-response bias is to minimize the number of non-responders—that is, to maximize the response rate. There is a large research literature on the factors that affect survey response rates (Groves et al., 2004)². In general, in-person interviews have the highest response rates, followed by telephone surveys, and then mail and Internet surveys. Among the other factors that increase response rates are sending potential respondents a short pre-notification message informing them that they will be asked to participate in a survey in the near future and sending simple follow-up reminders to non-responders after a few weeks. The perceived length and complexity of the survey can also make a difference, which is why it is important to keep survey questionnaires as short, simple, and on topic as possible. Finally, offering an incentive—especially cash—is a reliable way to increase response rates. However, ethically, there are limits to offering incentives that may be so large as to be considered coercive.

Conducting the Survey

The four main ways to conduct surveys are through in-person interviews, by telephone, through the mail, and over the internet. As with other aspects of survey design, the choice depends on both the researcher's goals and the budget. In-person interviews have the highest response rates and provide the closest personal contact with respondents. Personal contact can be important, for example, when the interviewer must see and make judgments about respondents, as is the case with some mental health interviews. But in-person interviewing is by far the most costly approach. Telephone surveys have lower response rates and still provide some personal contact with respondents. They can also be costly but are generally less so than in-person interviews. Traditionally, telephone directories have provided fairly comprehensive sampling frames. However, this trend is less true today as more people choose to only have cell phones and do not install land lines that would be included in telephone directories. Mail surveys are less costly still but generally have even lower response rates—making them most susceptible to non-response bias.

Not surprisingly, internet surveys are becoming more common. They are increasingly easy to construct and use (see “Online Survey Creation”). Although initial contact can be made by mail with a link provided to the

survey, this approach does not necessarily produce higher response rates than an ordinary mail survey. A better approach is to make initial contact by email with a link directly to the survey. This approach can work well when the population consists of the members of an organization who have known email addresses and regularly use them (e.g., a university community). For other populations, it can be difficult or impossible to find a comprehensive list of email addresses to serve as a sampling frame. Alternatively, a request to participate in the survey with a link to it can be posted on websites known to be visited by members of the population. But again it is very difficult to get anything approaching a random sample this way because the members of the population who visit the websites are likely to be different from the population as a whole. However, internet survey methods are in rapid development. Because of their low cost, and because more people are online than ever before, internet surveys are likely to become the dominant approach to survey data collection in the near future.

Finally, it is important to note that some of the concerns that people have about collecting data online (e.g., that internet-based findings differ from those obtained with other methods) have been found to be myths. Table 7.3 (adapted from Gosling, Vazire, Srivastava, & John, 2004)³ addresses three such preconceptions about data collected in web-based studies:

Table 7.3 Some Preconceptions and Findings Pertaining to Web-based Studies

Preconception	Finding
Internet samples are not demographically diverse	Internet samples are more diverse than traditional samples in many domains, although they are not completely representative of the population
Internet samples are maladjusted, socially isolated, or depressed	Internet users do not differ from nonusers on markers of adjustment and depression
Internet-based findings differ from those obtained with other methods	Evidence so far suggests that internet-based findings are consistent with findings based on traditional methods (e.g., on self-esteem, personality), but more data are needed.

Online Survey Creation

There are now several online tools for creating online questionnaires. After a questionnaire is created, a link to it can then be emailed to potential respondents or embedded in a web page. The following websites are among those that offer free accounts. Although the free accounts limit the number of questionnaire items and the number of respondents, they can be useful for doing small-scale surveys and for practicing the principles of good questionnaire construction. Here are some commonly used online survey tools:

- SurveyMonkey—<https://surveymonkey.com>
- PsyToolkit—<https://www.psychtoolkit.org/> (free, noncommercial, and does many experimental paradigms)
- Qualtrics—<https://www.qualtrics.com/>

- PsycData—<https://www.psychdata.com/>

A small note of caution: the data from US survey software are held on US servers, and are subject to be seized as granted through the Patriot Act. To avoid infringing on any rights, the following is a list of online survey sites that are hosted in Canada:

- Fluid Surveys—<http://fluidsurveys.com/>
- Simple Survey—<http://www.simplesurvey.com/>
- Lime Survey—<https://www.limesurvey.org>

There are also survey sites hosted in other countries outside of North America.

Another new tool for survey researchers is Mechanical Turk (MTurk) created by Amazon.com <https://www.mturk.com> Originally created for simple usability testing, MTurk has a database of over 500,000 workers from over 190 countries⁴. You can put simple tasks (for example, different question wording to test your survey items), set parameters as your sample frame dictates and deploy your experiment at a very low cost (for example, a few cents for less than 5 minutes). MTurk has been lauded as an inexpensive way to gather high-quality data (Buhrmester, Kwang, & Gosling, 2011)⁵.

Notes

1. Lahaut, V. M. H. C. J., Jansen, H. A. M., van de Mheen, D., & Garretsen, H. F. L. (2002). Non-response bias in a sample survey on alcohol consumption. *Alcohol and Alcoholism*, 37, 256–260.
2. Groves, R. M., Fowler, F. J., Couper, M. P., Lepkowski, J. M., Singer, E., & Tourangeau, R. (2004). *Survey methodology*. Hoboken, NJ: Wiley.
3. Gosling, S. D., Vazire, S., Srivastava, S., & John, O. P. (2004). Should we trust web-based studies? A comparative analysis of six preconceptions about internet questionnaires. *American Psychologist*, 59(2), 93–104.
4. Nataala@aws. (2011, January 26). Re: MTurk CENSUS: About how many workers were on Mechanical Turk in 2010? Message posted to Amazon Web Services Discussion Forums. Retrieved from <https://forums.aws.amazon.com/thread.jspa?threadID=58891>
5. Buhrmester, M., Kwang, T., & Gosling, S.D. (2011). Amazon's Mechanical Turk: A new source of inexpensive, yet high quality, data? *Perspectives on Psychological Science*, 6(1), 3–5.

37. Key Takeaways and Exercises

Key Takeaways

- Survey research features the use of self-report measures on carefully selected samples. It is a flexible approach that can be used to study a wide variety of basic and applied research questions.
- Survey research has its roots in applied social research, market research, and election polling. It has since become an important approach in many academic disciplines, including political science, sociology, public health, and, of course, psychology.
- Survey research involves asking respondents to self-report on their own thoughts, feelings, and behaviors.
- Most survey research is non-experimental in nature (it is used to describe variables or measure statistical relationships between variables) but surveys can also be used to measure dependent variables in true experiments.
- Responding to a survey item is itself a complex cognitive process that involves interpreting the question, retrieving information, making a tentative judgment, putting that judgment into the required response format, and editing the response.
- Survey responses are subject to numerous context effects due to question wording, item order, response options, and other factors. Researchers should be sensitive to such effects when constructing surveys and interpreting survey results.
- Survey items are either open-ended or closed-ended. Open-ended items simply ask a question and allow respondents to answer in whatever way they want. Closed-ended items ask a question and provide several response options that respondents must choose from.
- Use verbal labels instead of numerical labels although the responses can be converted to numerical data in the analyses.
- According to the BRUSO model, questionnaire items should be brief, relevant, unambiguous, specific, and objective.
- Survey research usually involves probability sampling, in which each member of the population has a known probability of being selected for the sample. Types of probability sampling include simple random sampling, stratified random sampling, and cluster sampling.
- Sampling bias occurs when a sample is selected in such a way that it is not representative of the population and therefore produces inaccurate results. The most pervasive form of sampling bias is non-response bias, which occurs when people who do not respond to the survey differ in important ways from people who do respond. The best way to minimize non-response bias is to maximize the response rate by prenotifying respondents, sending them reminders, constructing questionnaires that are short and easy to complete, and offering incentives.
- Surveys can be conducted in person, by telephone, through the mail, and on the internet. In-person interviewing has the highest response rates but is the most expensive. Mail and internet surveys are less expensive but have much lower response rates. Internet surveys are likely to become the dominant approach because of their low cost.

Exercises

- Discussion: Think of a question that each of the following professionals might try to answer using survey research.
 - a social psychologist
 - an educational researcher
 - a market researcher who works for a supermarket chain
 - the mayor of a large city
 - the head of a university police force
- Discussion: Write a survey item and then write a short description of how someone might respond to that item based on the cognitive model of survey responding (or choose any item on the Rosenberg Self-Esteem Scale at <http://www.bsos.umd.edu/socy/research/rosenberg.htm>).
- Practice: Write survey items for each of the following general questions. In some cases, a series of items, rather than a single item, might be necessary.
 - How much does the respondent use Facebook?
 - How much exercise does the respondent get?
 - How likely does the respondent think it is that the incumbent will be re-elected in the next presidential election?
 - To what extent does the respondent experience “road rage”?
- Discussion: If possible, identify an appropriate sampling frame for each of the following populations. If there is no appropriate sampling frame, explain why.
 - students at a particular university
 - adults living in the state of Washington
 - households in Pullman, Washington
 - people with low self-esteem
- Practice: Use one of the online survey creation tools to create a 10-item survey questionnaire on a topic of your choice.